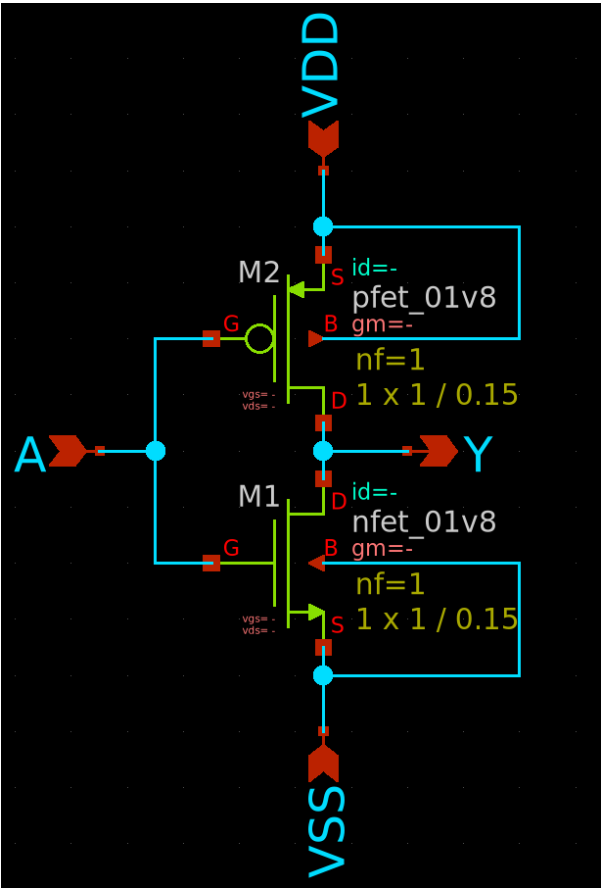


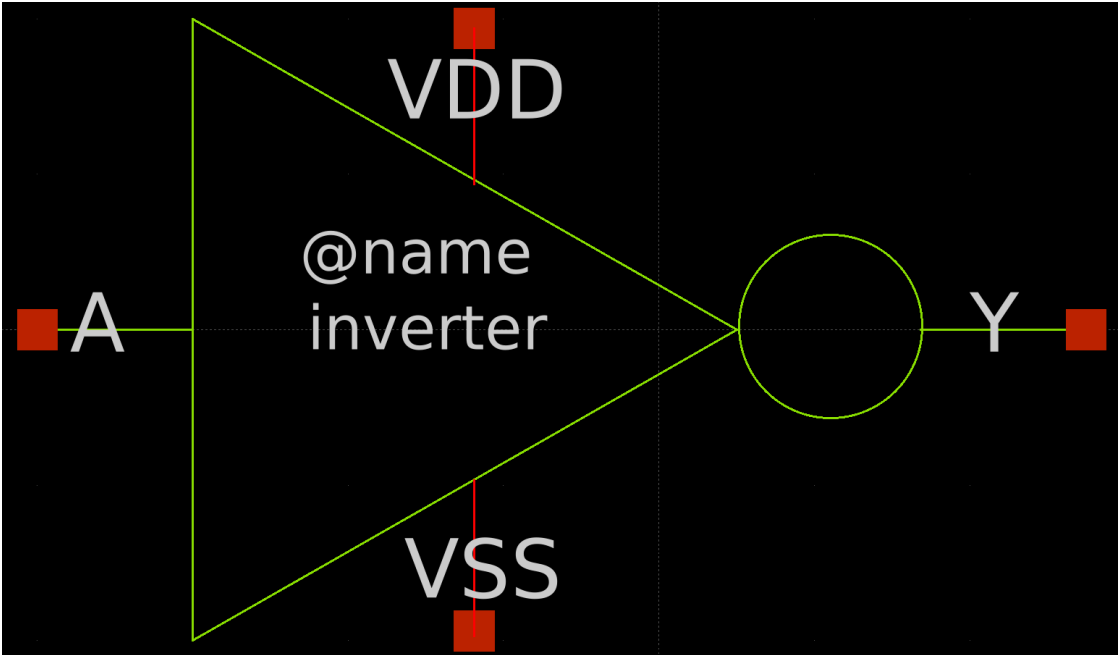
Familiarisation of IIC-OSIC-TOOLS

Deva Prassad S

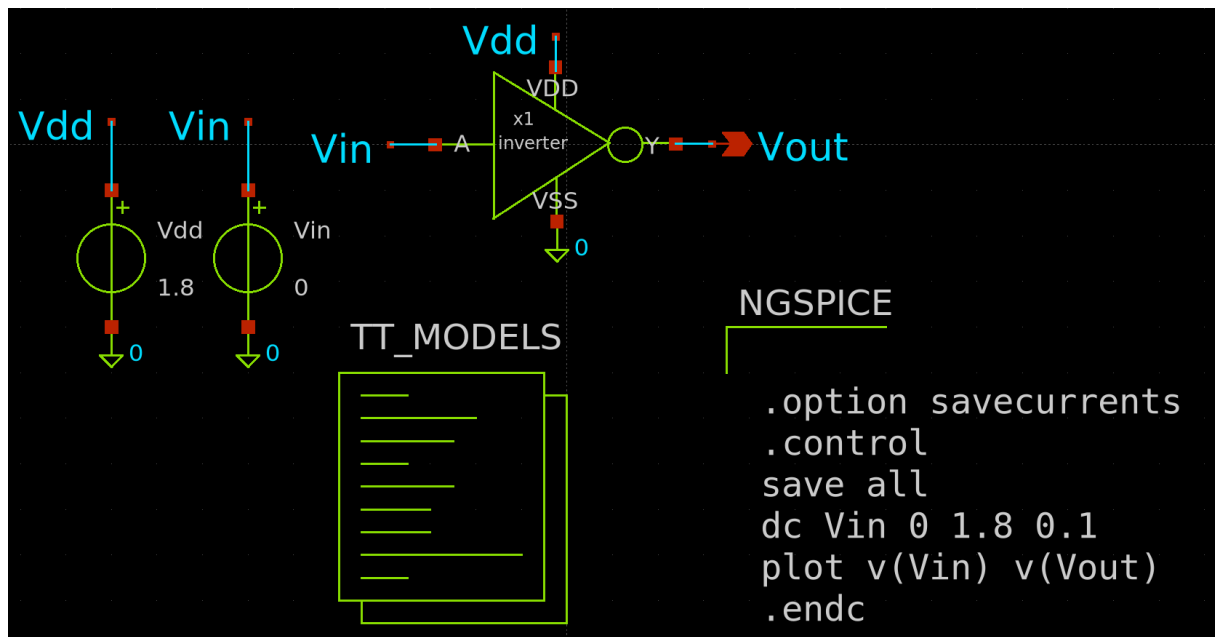
Schematic Capture



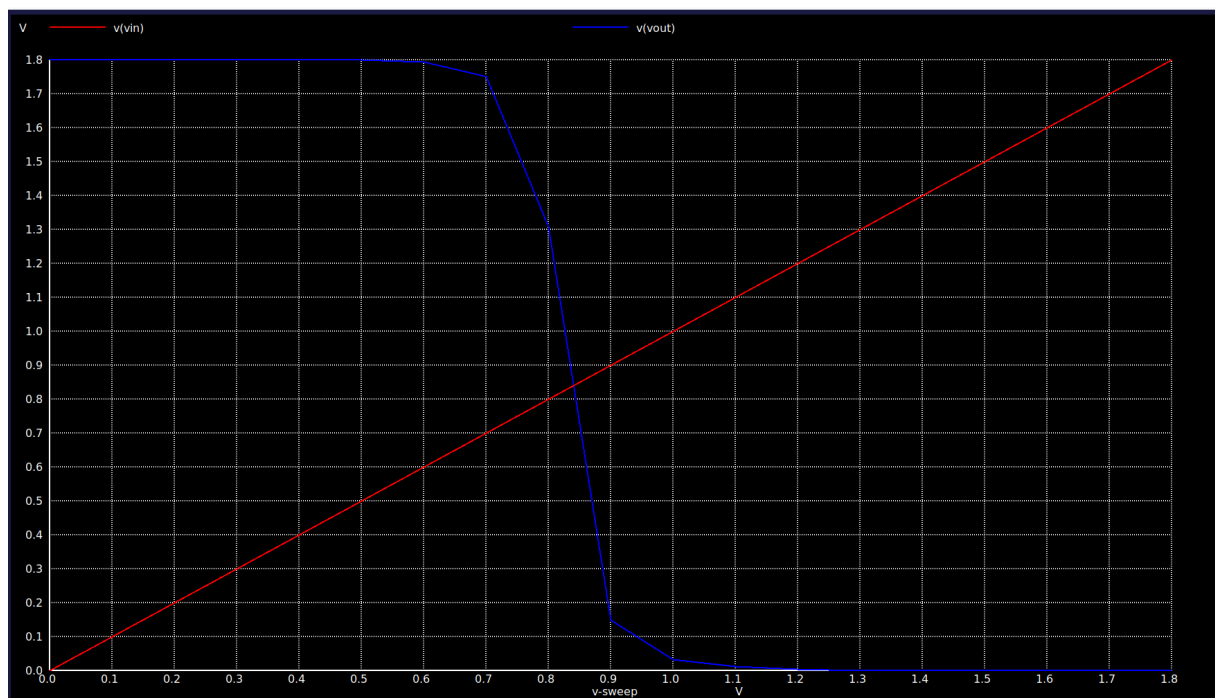
Symbol Creation



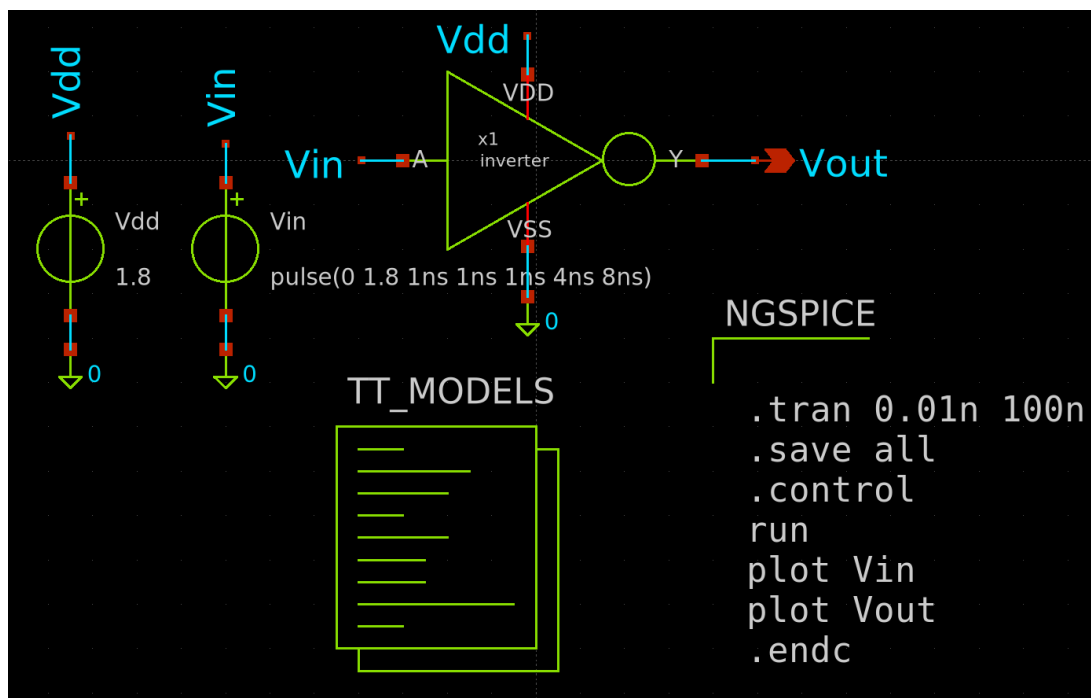
DC Test Bench



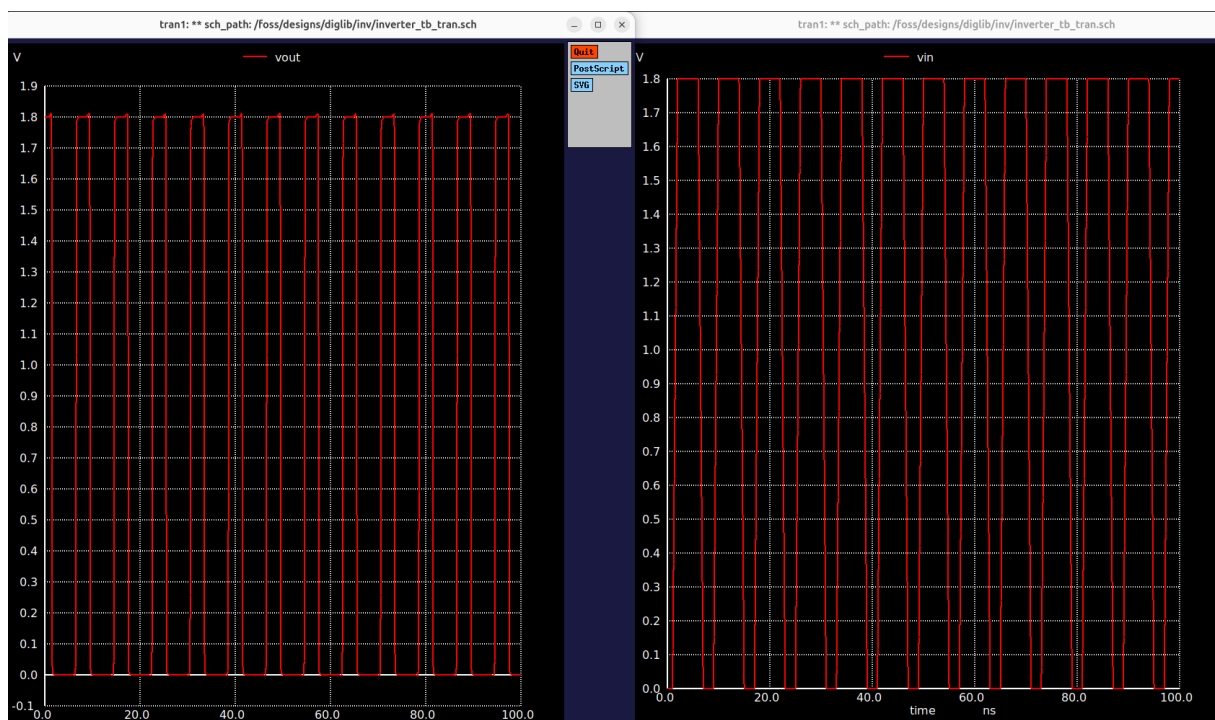
DC Result — Voltage Transfer Characteristic



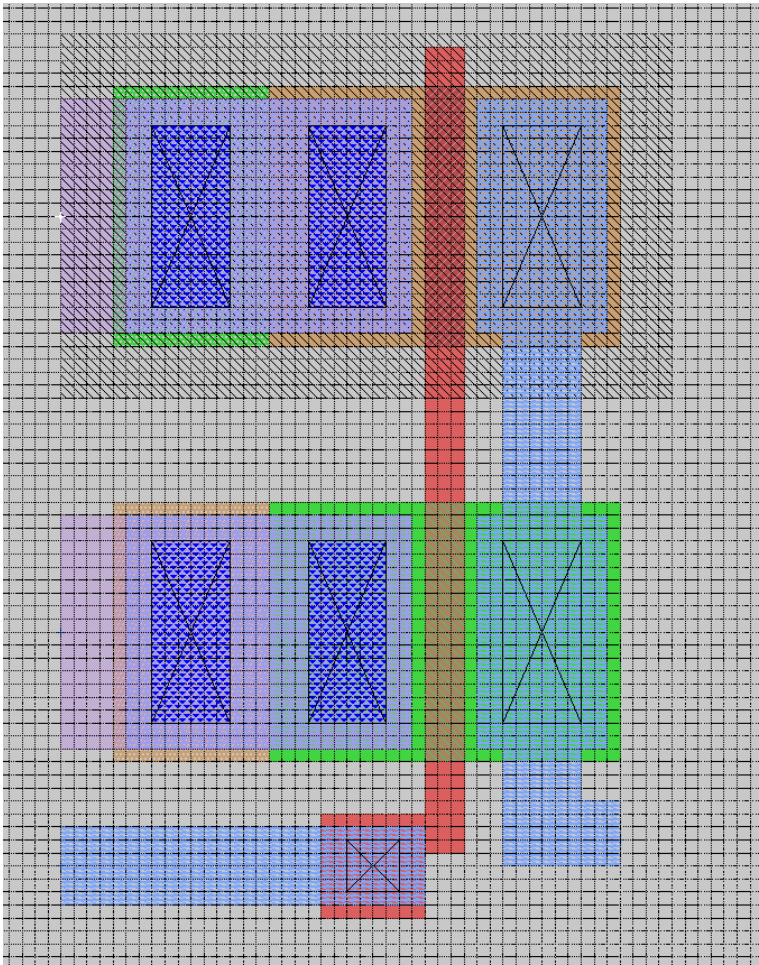
Transient Test Bench



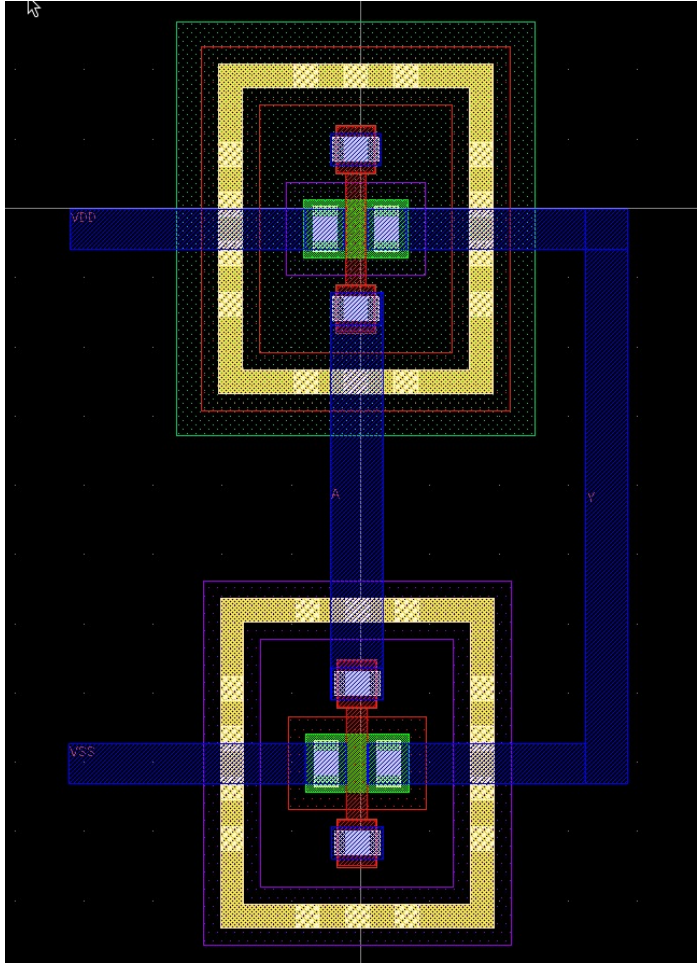
Transient Result



Layout — Magic



Layout — KLayout View



Design Rule Check (DRC)

```
/foss/designs/simulations/inverter > iic-drc.sh
```

DRC script for Magic and KLayout (ICD@JKU)

Usage: /foss/tools/sak/iic-drc.sh [-d] [-m|-k|-b] [-c] [-l <level>] [-f <pattern>] [-w <workdir>] <cellname>

- m Run Magic DRC (default)
- k Run KLayout DRC
- b Run Magic and KLayout DRC
- c Clean output files before running
- l Klayout DRC level: precheck|macro|regular (default: macro)
 - precheck: core FEOL and BEOL rules only
 - macro: adds off-grid, pin, and zero-area checks, skips chip-level density and antenna
 - regular: all checks
- f Set GDS flatglob pattern for Magic (e.g. '*' to flatten all)
- w Use <workdir> to store result files (default current dir)
- d Enable debug information

```
/foss/designs/simulations/inverter > iic-drc.sh -d -b inverter.mag  
[INFO] DEBUG is enabled!  
[INFO] flag -b is set.  
[INFO] sky130 PDK selected.  
[INFO] Using layout file <inverter.mag>.br/>[WARNING] KLayout DRC needs a GDS layout, running Magic DRC only.  
[INFO] Running DRC of <inverter.mag>.br/>[INFO] Results are put into </foss/designs/simulations/inverter>.br/>[INFO] Launching Magic DRC...  
[INFO] Magic runs DRC on the .mag file.
```

```
[INFO] Magic DRC is clean!
```

CONGRATULATIONS! No DRC errors in <inverter> found!

"

"

```
[DONE] Bye!  
/foss/designs/simulations/inverter > 
```

Layout-vs-Schematic (LVS)

```
/foss/designs/simulations/inverter > iic-lvs.sh

LVS script for Magic+Netgen and KLayout (ICD@JKU)

Usage: /foss/tools/sak/iic-lvs.sh [-d] [-m|-k|-b] [-w <workdir>] [-s <schematic>|-<netlist> -l <layout> -c <cellname> ] <cellname>

Specify <cellname> to use for schematic and layout, resolved against the
current dir using default name prefixes (.sch|.spice|.spc|.cdl|.v|.mag|
.mag.gz|.gds|.gds.gz|.klay.gds). When no <cellname> is specified use -s,
-l, and -c to point to the corresponding files and name the topcell.

-m Run Magic+Netgen LVS (default)
-k Run KLayout LVS
-b Run Magic+Netgen and KLayout LVS
-s Use this <schematic> (xschem .sch schematic, SPICE/CDL netlist, or Verilog netlist)
-l Use this <layout> view
-c Name of <topcell>
-w Use <workdir> to store result files (default current dir)
-d Enable debug information

/foss/designs/simulations/inverter > iic-lvs.sh -d -s inverter.sch -l inverter.mag -c inverter
[INFO] DEBUG is enabled!
[INFO] flag -s is set to <inverter.sch>.
[INFO] flag -l is set to <inverter.mag>.
[INFO] flag -c is set to <inverter>.
[INFO] sky130 PDK selected.
[INFO] Using schematic file </foss/designs/simulations/inverter/inverter.sch>.
[INFO] Using layout file </foss/designs/simulations/inverter/inverter.mag>.
[INFO] Running LVS of </foss/designs/simulations/inverter/inverter.mag> vs </foss/designs/simulations/inverter/inverter.sch>.
[INFO] Extracting SPICE netlist from schematic </foss/designs/simulations/inverter/inverter.sch>...
[INFO] Extracting netlist from layout </foss/designs/simulations/inverter/inverter.mag>...
[INFO] Run netgen...
---
[INFO] Magic+Netgen LVS is OK, schematic/netlist and layout match!
---
CONGRATULATIONS! LVS is OK, schematic/netlist and layout match!

      *****
      (o) (o)
      <
      \___/
      *****
---
[DONE] Bye!
/foss/designs/simulations/inverter > 
```

Parasitic Extraction (PEX)

```
/foss/designs/simulations/inverter > iic-pex.sh

PEX script using Magic (ICD@JKU)

Usage: /foss/tools/sak/iic-pex.sh [-d] [-m <mode>] [-s <mode>] [-n <subcktname>] [-w <workdir>]
      [-t <threshold>] [-r <minres>] [-y <mindelay>] <cellname>

-m Select PEX mode (1 = C-decoupled, 2 = C-coupled [default], 3 = full-RC)
-s Subcircuit definition in PEX netlist (1 = include subcircuit definition [default], 0 = no subcircuit)
-n Name of PEX subcircuit (default is <cellname>)
-w Use <workdir> to store result files (default current dir)
-t full-RC only: extresist threshold in mOhm (default 10000 = 10 Ohm)
-r full-RC only: extresist minres in mOhm (default 1000 = 1 Ohm)
-y full-RC only: extresist mindelay in ps (default 1, 0 = gate by resistance instead of delay)
-d Enable debug information

<cellname> may be a cell name or a layout file (.mag, .mag.gz, .gds, .gds.gz)
NOTE: for a GDS input the top cell must be named like the file (<cellname>)
```

```
/foss/designs/simulations/inverter > iic-pex.sh -d -m 3 inverter.mag
```

```
[INFO] DEBUG is enabled!
[INFO] flag -m is set to <3>.
[INFO] sky130 PDK selected.
[INFO] Using layout file <inverter.mag>.
[INFO] Running PEX of <inverter.mag>.
[INFO] Results are put into </foss/designs/simulations/inverter>.
```

Magic 8.3 revision 678 - Compiled on Mon Jul 27 03:07:22 PM CEST 2026.

Starting magic under Tcl interpreter

Using the terminal as the console.

WARNING: RLIMIT_NOFILE is above 1024 and Tcl_Version<9 this may cause runtime issues [rlim_cur=1048576]

Using NULL graphics device.

Processing system .magicrc file

Switching to WIRING tool.

Switching to NETLIST tool.

Switching to PICK tool.

Switching to BOX tool.

Sourcing design .magicrc for technology sky130A ...

2 Magic internal units = 1 Lambda

Input style sky130(): scaleFactor=2, multiplier=2

The following types are not handled by extraction and will be treated as non-electrical types:

ubm

Scaled tech values by 2 / 1 to match internal grid scaling

Loading sky130A Device Generator Menu ...

Loading "/foss/designs/simulations/inverter/pex_inverter.tcl" from command line.

Cell inverter read from current working directory

Scaled magic input cell inverter geometry by factor of 2

Extracting inverter into /foss/designs/simulations/inverter/inverter.ext:

Processing inverter

Processing cell inverter for resistance extraction.

Extracting VSS

Total Nets: 4

Nets extracted: 1 (0.250000)

Nets output: 1 (0.250000)

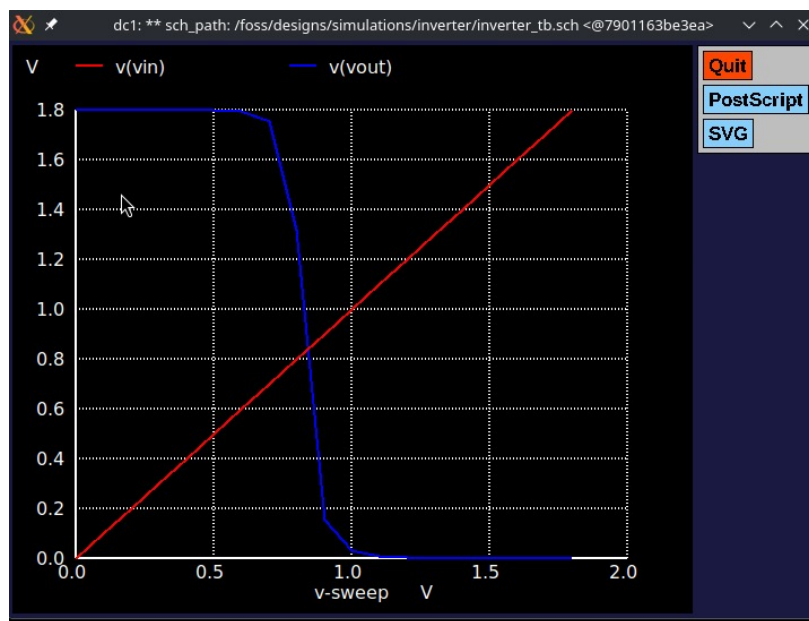
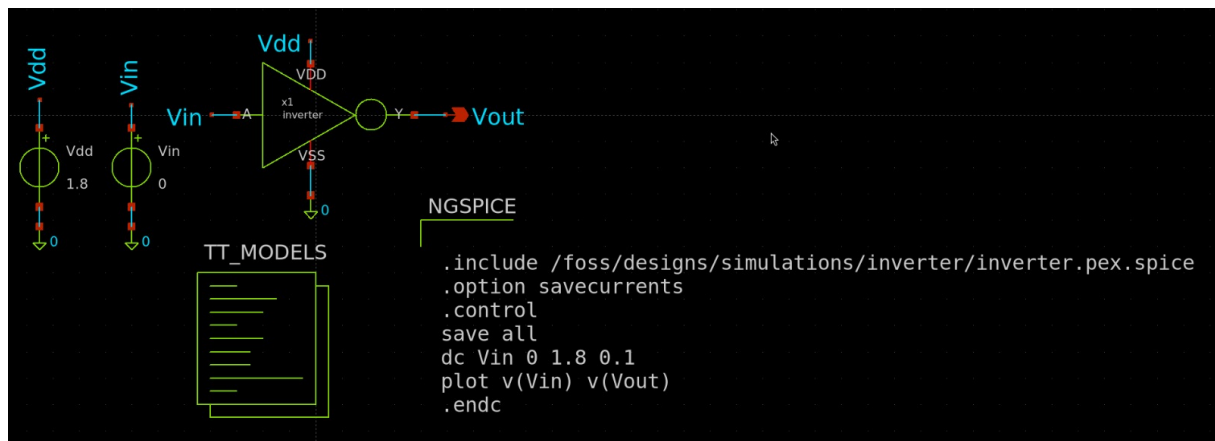
exttospcice finished.

[INFO] PEX (full-RC) done, the extracted SPICE netlist is </foss/designs/simulations/inverter/inverter.pex.spice>.

[DONE] Bye!

```
/foss/designs/simulations/inverter > █
```

Post-Layout DC



Post-Layout Transient

